

- Each statement below concerns one part of a statement of the Midterm collection, and it may be worded differently from the collection. Clearly **select** (a) if the statement is true and (b) if it is false. Marks are +1 for a correct answer, −1 for a wrong answer, and 0 for no answer.
- No derivations or justifications are required, and none is marked.
- The exam **grade** is  $R - W$ , the number of right answers minus the number of wrong answers, over the 12 statements.
- No electronic devices and no printed or written material are allowed.
- The exam duration is **30 minutes**.

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**Name:**

**NIA or U:**

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1. The inner product of two finite-energy waveforms  $x(t)$  and  $y(t)$  is

$$\langle x, y \rangle = \int_{-\infty}^{\infty} x(t)y^*(t) dt,$$

with the complex conjugate on the second waveform.

- (a) True
  - (b) False
2. If a waveform  $\phi(t)$  has energy  $E$ , then the time-scaled waveform  $\phi(2t)$  has energy  $2E$ .
- (a) True
  - (b) False
3. The waveform  $\text{sinc}(t)$  has unit energy and its Fourier transform occupies the band  $|f| \leq 1$ .
- (a) True
  - (b) False
4. A baseband waveform of bandwidth  $W$ , which occupies  $|f| \leq W$ , allows at most  $2W$  symbols per second without inter-symbol interference.
- (a) True
  - (b) False
5. A 256-QAM constellation carries 8 bits per symbol, so symbols sent at  $R_s$  symbols per second give a bit rate of  $8R_s$  bits per second.
- (a) True
  - (b) False
6. The energy of the sum of two signals equals the sum of their energies, whatever the two signals are.
- (a) True
  - (b) False
7. In MATLAB, the expression `2*rand + 1` returns a random number that is uniform on the interval  $(1, 2)$ .
- (a) True
  - (b) False
8. The entropy of a discrete random variable that takes  $M$  values is at most  $\log_2 M$  bits, and it equals  $\log_2 M$  bits when the  $M$  values are equiprobable.
- (a) True
  - (b) False
9. Among all real random variables with the same variance, the uniform random variable has the largest differential entropy.
- (a) True
  - (b) False
10. The capacity of the binary symmetric channel with crossover probability  $p$  vanishes at  $p = 1$ .
- (a) True
  - (b) False

11. The capacity of the complex AWGN channel with average symbol energy  $E_s$  and noise variance  $\sigma^2$  is

$$C = \log_2 \left( 1 + \frac{E_s}{\sigma^2} \right)$$

bits per channel use, and it is achieved by a Gaussian input.

- (a) True
  - (b) False
12. When the symbols are equiprobable, the maximum likelihood detector and the maximum a posteriori detector make the same decisions.
- (a) True
  - (b) False